

ORF307: Optimization

Precept: #7

Agenda

- Review of the Simplex method
- Optimality condition question
- Degeneracy and cycling
- Example

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Standard form of polyhedra

Throughout this precept we assume we are interested in a **minimization LP**.

Recall that a polyhedron is defined as the intersection of half-spaces

$$P = \{x \in \mathbb{R}^n \mid a_i^T x \leq b_i, \ i = 1, 2, \dots, m\}$$

where $a_i, x \in \mathbb{R}^n$. It can also be defined in a more compact form as

$$P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$

We can convert the above polyhedron to an equivalent form where the affine inequalities have been converted to affine equalities, by adding slack variables:

$$P = \{(x, s) \in \mathbb{R}^{n+m} \mid a_i^T x + s_i = b_i, \ s_i \geq 0, \ i = 1, 2, \dots, m\}$$

or in compact form

$$P = \{(x, s) \in \mathbb{R}^{n+m} \mid Ax + s = b, \ s \geq 0\}$$

Remark: This definition of the polyhedron (with the addition of the slack variables) provides an easy way to find an **initial basic feasible solution** (BFS). If $b \geq 0$, we can

set the original variables $x = 0$ (non-basic) and the slack variables $s = b$ (basic), yielding the BFS: $(x, s) = (0, b) \geq 0$. On the other hand, if $b < 0$ we only obtain a basic solution (BS).

We say that a polyhedron is in **standard form** if it is defined as follows:

$$P = \{x \in \mathbb{R}^n | Ax = b, x \geq 0\}$$

Note that in this definition we do **not** have slack variables. All decision variables must be nonnegative and the constraints must be equalities.

Remark: In this final form, all variables (original, slack, surplus, and artificial) are consolidated into the single **non-negative** vector x . This allows us to work with **a single constraint matrix** A and **a unified set of non-negativity constraints** $x \geq 0$, which *simplifies* the definition of the basis B and the calculation of reduced costs.

Feasible directions

Consider a polyhedron in **standard form**:

$$P = \{x \in \mathbb{R}^n | Ax = b, x \geq 0\} \tag{1}$$

A vector $d \in \mathbb{R}^n$ is a **feasible direction at** $x \in P$ if and only if

$$\exists \theta > 0 \text{ such that } x + \theta d \in P$$

CLAIM: If d is a feasible direction then $Ad = 0$.

Proof: Since d is a feasible direction, $\exists \theta > 0$ *sufficiently small*, such that for any $x \in P$ we also have $x + \theta d \in P$. This means: $x + \theta d \geq 0$ **and** $A(x + \theta d) = b$.

Then we have:

$$A(x + \theta d) = b \iff Ax + \theta Ad = b \iff \theta Ad = 0 \iff Ad = 0$$

Remark: The CLAIM holds only when the feasible region is described by a polyhedron (which defines the feasible region) that is in standard form.

Remark: Note the distinct roles of the non-negativity constraints ($x \geq 0$) and the equality constraints ($Ax = b$), when determining a feasible direction (d). The equality $Ad = 0$ directly arises from the requirement that $x + \theta d$ **must satisfy** the equality constraints $Ax = b$. On the other hand, the non-negativity constraints $x + \theta d \geq 0$ are used to determine the appropriate **step-size** θ that guarantees non-negativity of all decision variables in the next iterate (we will discuss this in more detail in future precepts).

Basic Feasible Solution (BFS)

Let us consider again a polyhedron in standard form.

$$P = \{x \in \mathbb{R}^n | Ax = b, x \geq 0\}$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ and $x \in \mathbb{R}^n$.

- We define the set of indices of the columns of A as

$$J = \{1, 2, \dots, n\}$$

- A basis B is a subset of the column set J (i.e., $B \subseteq J$), such that:
 1. the set B contains **exactly** m elements (that is the cardinality of the set B equals m , or in mathematical notation: $|B| = m$),
 2. if we denote by $B(1), B(2), \dots, B(m)$ the column indices in the basis B , i.e.,

$$B = \{B(1), B(2), \dots, B(m)\}$$

then the corresponding columns $A_{B(1)}, A_{B(2)}, \dots, A_{B(m)}$ of the matrix A are **linear independent**. This means that the $m \times m$ matrix (square matrix)

$$A_B = [A_{B(1)}, A_{B(2)}, \dots, A_{B(m)}]$$

is **invertible** (i.e., non-singular). We can also deduce that $\text{rank}(A_B) = m$.

- The variables corresponding to the indices in B are called **basic variables**.
- All other variables are called **non-basic variables**. We can denote by N the set of nonbasic indices. Then we have

$$B \cup N = J \text{ and } B \cap N = \emptyset$$

- $x \in \mathbb{R}^n$ is called a **basic solution** if

$$A_B x_B = b \quad \text{and} \quad x_i = 0, \quad \forall i \notin B \quad (\text{or } \forall i \in N)$$

Note that at this point we do **not** know whether all elements of x_B are non-negative.

- $x \in \mathbb{R}^n$ will be called **Basic Feasible Solution** if it is basic solution **and** $x_B \geq 0$. In this case we have $Ax = b$ and $x \geq 0$, since

$$Ax = b \underset{x = (x_B, x_N)}{\iff} A_B x_B + A_N x_N = b \underset{x_N = 0}{\iff} A_B x_B = b \underset{\exists A_B^{-1}}{\iff} x_B = A_B^{-1} b \geq 0$$

Basic feasible direction d

- The j -th **basic direction** means we are considering adding the *non-basic index* j (i.e., $j \notin B$) to the basis B .
- Remember, we need to have $Ad = 0$ and we do not want to add other non-basic variables to the basis
- The procedure of constructing the j -th basic direction is:
 - Set $d_j = 1$, for some $j \notin B$ (equivalently, for some $j \in N$)

- Set $d_k = 0$ for all $k \neq j$ and $k \notin B$ (equivalently, $k \neq j$ and $k \in N$)
- Then compute d_B by solving the linear system $Ad = 0$. To see how let's **expand** the system:

$$\begin{aligned}
Ad = 0 &\iff \sum_{i=1}^n A_i d_i = 0 \iff \sum_{i \in B} A_i d_i + \sum_{i \in N} A_i d_i = 0 \iff \\
&\iff_{\substack{(d_j = 1 \text{ and} \\ d_k = 0, \forall k \in N, k \neq j)}} A_B d_B + A_j d_j = 0 \iff A_B d_B = -A_j \iff \\
&\iff d_B = -A_B^{-1} A_j
\end{aligned}$$

Cost of improvement

- We can easily find by how much the cost function has improved by moving along the basic feasible direction d :

$$\underset{\text{new cost}}{c^T(x + \theta d)} - \underset{\text{old cost}}{c^T x} = \underset{\text{cost improvement}}{\theta c^T d}$$

- $c^T d$ is the **reduced cost** associated with the feasible direction d (and step-size $\theta = 1$).
- The change in the objective by adding the new variable x_j to the basis is defined by

$$\begin{aligned}
\bar{c}_j = c^T d &= \sum_{i=1}^n c_i d_i = \sum_{i \in N} c_i d_i + \sum_{i \in B} c_i d_i = \\
&= c_j + c_B^T d_B \underset{(d_B = -A_B^{-1} A_j)}{=} c_j - c_B^T A_B^{-1} A_j
\end{aligned}$$

Remark: $\bar{c}_j$ is the reduced cost associated with the non-basic variables x_j . It tells us the **rate of change** of the objective function $c^T x$ as we increase x_j from 0 to some non-zero value (and simultaneously adjust all basic variables x_B according to direction d_B).

Remark: If we decide to make another non-basic variable, say x_ℓ , to basic, then we set

$$d_\ell = 1, \text{ and } d_k = 0, \forall k \neq \ell, \text{ and } d_B = -A_B^{-1} A_\ell$$

We can similarly compute the reduced cost $\bar{c}_\ell$ and the change in the objective when we add the new variable x_ℓ as:

$$\bar{c}_\ell = c_\ell - c_B^T A_B^{-1} A_\ell$$

- **Remark:** The reduced cost vector $\bar{c}$ contains the reduced costs for all non-basic variables.
- **Termination criterion:** if $\bar{c} \geq 0$ then x is **Optimal** and we **Stop**.

That is, if the reduced costs associated with all non-basic variables are non-negative, then we have reached the optimal vertex and we cannot improve the objective any further (and we Stop).

- If there is some $\bar{c}_j < 0$ then we can decrease the cost function by **bringing** x_j **in into the basis**.

Remark: if we had a *maximization problem*, the termination criteria *reverse* (i.e., if $\bar{c} \leq 0$ then x is optimal (maximum) and we Stop).

If there is a $\bar{c}_j > 0$ then we can increase the cost by bring the corresponding j into the basis.

- **Step-size:** once the direction d is determined, we have to compute the largest possible step-size θ^* along that direction, such that the next iterate, $x + \theta^*d$, is still no-negative, that is

$$\theta^* = \max\{\theta \in \mathbb{R} \mid \theta \geq 0, x + \theta d \geq 0\}$$

There is an explicit formula that easily determines the maximum step-size θ^* and is based on simple computations. Let's see how we can derive that formula:

- For all indices i such that $d_i > 0$ and for all $\theta \geq 0$, we will have $x_i + \theta d_i \geq 0$, because x is feasible (and hence $x \geq 0$).
- However, we have to be very careful when there is an index such that $d_i < 0$, because in that case $x_i + \theta d_i$ can become negative (infeasible), if θ is chosen as a large positive number. Then the **largest allowable step-size** θ_i that we can choose for the variable x_i when $d_i < 0$ is the one that makes $x_i + \theta_i d_i = 0$, that is

$$x_i + \theta_i d_i = 0 \implies \theta_i = \frac{-x_i}{d_i}$$

Then θ^* is selected as the **smallest** of all those θ_i 's:

$$\theta^* = \min_{\{i \in B \mid d_i < 0\}} \left\{ \frac{-x_i}{d_i} \right\}$$

The step-size θ^* computed by the above formula **guarantees**: $x + \theta^*d \geq 0$.

Remark: the formula we use for determining the step size θ^* is the **same** for both minimization and maximization problems.

- If $d \geq 0$ (that is $d_i \geq 0, \forall i$) then for any large $\theta > 0$ we can increase the objective function arbitrarily, and therefore the LP is **unbounded**.

Optimality conditions

Theorem. A feasible solution $x \in P$ is optimal (minimum) for the objective function $c^T x$ over a convex feasible set P if and only if $c^T d \geq 0$ for every feasible direction d at x .

Proof. We first prove the necessary ($\implies$) condition and then the sufficient ($\impliedby$) one.

($\implies$) Assume that $x \in P$ is an optimal solution (minimum). We want to show that for any feasible direction d , we must have $c^T d \geq 0$ (i.e., the termination criterion is satisfied).

Suppose, for contradiction, that there exists a feasible direction d at x such that $c^T d < 0$. By the definition of a feasible direction, there must exist some $\bar{\theta} > 0$ such that the point $x + \theta d$ is in the feasible set P for all $\theta \in [0, \bar{\theta}]$.

Consider the objective function value at the new feasible point $x + \theta d$:

$$c^T(x + \theta d) = c^T x + \theta(c^T d)$$

Since we assumed $c^T d < 0$ and $\theta > 0$, the term $\theta(c^T d)$ is strictly negative. This implies that:

$$c^T(x + \theta d) < c^T x$$

Thus, we found a feasible solution $(x + \theta d) \in P$ with a strictly lower objective function value than x . This **contradicts** the assumption that x is an optimal (minimum) solution. Therefore, our initial assumption must be false, and it must hold that $c^T d \geq 0$ for any feasible direction d at x .

($\Leftarrow$) Assume that $c^T d \geq 0$ for every feasible direction d at $x \in P$. We want to show that x is also an optimal (minimum) solution. That is, we need to show that $c^T x \leq c^T y$ for any other feasible solution $y \in P$.

We construct the direction vector $d = y - x$. Since P is a **convex set**, the line segment connecting x and y lies entirely within P . That is, for any $\theta \in [0, 1]$, the point $x + \theta d = x + \theta(y - x) = (1 - \theta)x + \theta y$ is in P . This means that $d = y - x$ is a **feasible direction** at x .

By our assumption we know that $c^T d \geq 0$, and substituting $d = y - x$ yields:

$$c^T(y - x) \geq 0 \iff c^T y - c^T x \geq 0 \iff c^T x \leq c^T y$$

Since y was an arbitrary point in P , this inequality holds for all $y \in P$, which confirms that x is an optimal (minimum) solution. $\square$

Remark. For a *maximization* problem with the objective function $c^T x$, the necessary and sufficient condition for a feasible solution $x \in P$ to be optimal is that $c^T d \leq 0$ for every feasible direction d at x . Also, the directions of the inequalities in the proof **reverse**.

General Remark: Throughout this Precept we considered the feasible to be defined by the polyhedron (1) in **standard form**. This allowed us to prove the **CLAIM**: "if d is a feasible direction then $Ad = 0$ ".

The CLAIM does NOT apply to general LPs with inequalities. That's why it is important to convert a general LP in standard form, because it allows us to

- use linear algebra tools (the null space of A) to describe feasible directions
- construct basic feasible directions, in a systematic way, by solving the smaller system of linear equations: $A_B d_B = -A_j$
- and, apply optimality conditions like $c^T d \geq 0$ for all feasible directions.

Example: Simplex iteration (also known as Simplex pivot)

Let's see in detail the operations performed in a single iteration of the Simplex method.

We consider the minimization problem:

$$\min c^T x = -3x_1 - x_2$$

subject to:

$$\begin{aligned}x_1 + x_2 &\leq 4 \\2x_1 + x_2 &\leq 5 \\x_1, x_2 &\geq 0\end{aligned}$$

Step #1 : Convert it into Standard Form.

We introduce two slack variables x_3, x_4 to convert inequalities into equalities. As a result two new variables and two new non-negativity constraints are added and the original LP takes the standard form:

$$\min c^T x = -3x_1 - x_2 + 0x_3 + 0x_4$$

subject to:

$$\begin{aligned}x_1 + x_2 + x_3 &= 4 \\2x_1 + x_2 + x_4 &= 5 \\x_1, x_2, x_3, x_4 &\geq 0\end{aligned}$$

Let:

- The new decision variables: $\tilde{x} = (x_1, x_2, x_3, x_4)$, which include the slacks.
- The new cost vector: $c = (-3, -1, 0, 0)$, which includes the coefficients for the slacks. We reuse (with a slight abuse of notation) the same symbol for convenience, avoiding the introduction of a new vector.
- Initial basic variables: x_3, x_4 and $B = \{3, 4\}$.
- Initial non-basic variables: x_1, x_2 and $N = \{1, 2\}$.
- Initial basic feasible solution: $x_1 = x_2 = 0$, and $x_3 = 4, x_4 = 5$. Hence

$$\tilde{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 4 \\ 5 \end{pmatrix}$$

- The value of the objective function at the initial BFS $\tilde{x}$ becomes

$$c^T \tilde{x} = 0$$

- **Remark:** note how easy it is to determine an initial BFS when we add slack variables and convert the the original LP into its equivalent standard form.

Step #2 : Identify Entering Variable

Compute reduced costs for $j \in N = \{1, 2\}$:

$$\bar{c}_j = c_j - c_B^T A_B^{-1} A_j$$

where

$$c_1 = -2, \quad c_2 = -1, \quad A_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad A_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

We find the following reduced costs:

$$\bar{c}_1 = -3, \quad \bar{c}_2 = -1$$

We choose x_1 to **enter the basis** (i.e., the most negative reduced cost).

Step #3 : Compute the **feasible direction vector** d

Solve:

$$A_B d_B = -A_j \quad \text{where } j = 1$$

Then:

$$d_B = -A_B^{-1} A_1 = -A_1 = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$$

Since the non-basic variable will enter the basis we set $d_1 = 1$ and $d_2 = 0$.

The complete direction vector is defined as:

$$d = (1, 0, -1, -2)^T$$

Step #4 : Compute the **step-size** θ^*

Use the ratio test:

$$\theta^* = \min_{\{i \in B \mid d_i < 0\}} \left\{ -\frac{x_i}{d_i} \right\}$$

Current basic variables:

- $x_3 = 4, \quad d_3 = -1 < 0 \Rightarrow \theta_3 = -\frac{4}{-1} = 4$
- $x_4 = 5, \quad d_4 = -2 < 0 \Rightarrow \theta_4 = -\frac{5}{-2} = 2.5$

The minimum step-size is

$$\theta^* = \min\{\theta_3, \theta_4\} = \min\{4, 2.5\} = 2.5$$

and corresponds to the variable x_4 , which **will leave the basis**.

Step #5 : **Solution update**

New solution:

$$x^{\text{new}} = \tilde{x} + \theta^* d \implies \begin{pmatrix} x_1^{\text{new}} \\ x_2^{\text{new}} \\ x_3^{\text{new}} \\ x_4^{\text{new}} \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + \theta^* \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix} \implies$$

$$\Rightarrow \begin{pmatrix} x_1^{\text{new}} \\ x_2^{\text{new}} \\ x_3^{\text{new}} \\ x_4^{\text{new}} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 4 \\ 5 \end{pmatrix} + 2.5 \begin{pmatrix} 1 \\ 0 \\ -1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2.5 \\ 0 \\ 1.5 \\ 0 \end{pmatrix}$$

The **pivot** is:

- x_1 enters the basis (becomes basic variable)
- x_4 leaves the basis (becomes non-basic variable)

Summary of the Simplex iteration (pivot)

Variable	Status	Value
x_1	Became Basic	2.5
x_2	Remained Non-basic	0
x_3	Remained Basic	1.5
x_4	Became Non-basic	0

The objective value at x^{new} is **smaller** than at the initial point x :

$$c^T x^{\text{new}} = -3 \times 2.5 - 1 \times 0 = -7.5$$

Remark: Note that the objective value has **decreased**. At the initial BFS $\tilde{x}$ we had $c^T \tilde{x} = 0$ and the new BFS we have $c^T x^{\text{new}} = -7.5$

Figure 1 shows the visualization of the pivot.

Remark: If you compute the reduced costs in the new BFS x^{new} you will see that they are non-negative. This means this vertex is the optimal solution. Please verify that by computing the reduced costs.

Practice exercise

Compute the optimal solution of the LP by performing 3 Simplex iterations (pivots).

$$\min c^T x = -3x_1 - x_2$$

subject to:

$$x_1 + x_2 \leq 100$$

$$x_2 \leq 40$$

$$x_1 - x_2 \leq 30$$

$$x_2 - x_1 \leq 30$$

$$x_1, x_2 \geq 0$$

Figure 2 shows the feasible region of the LP.

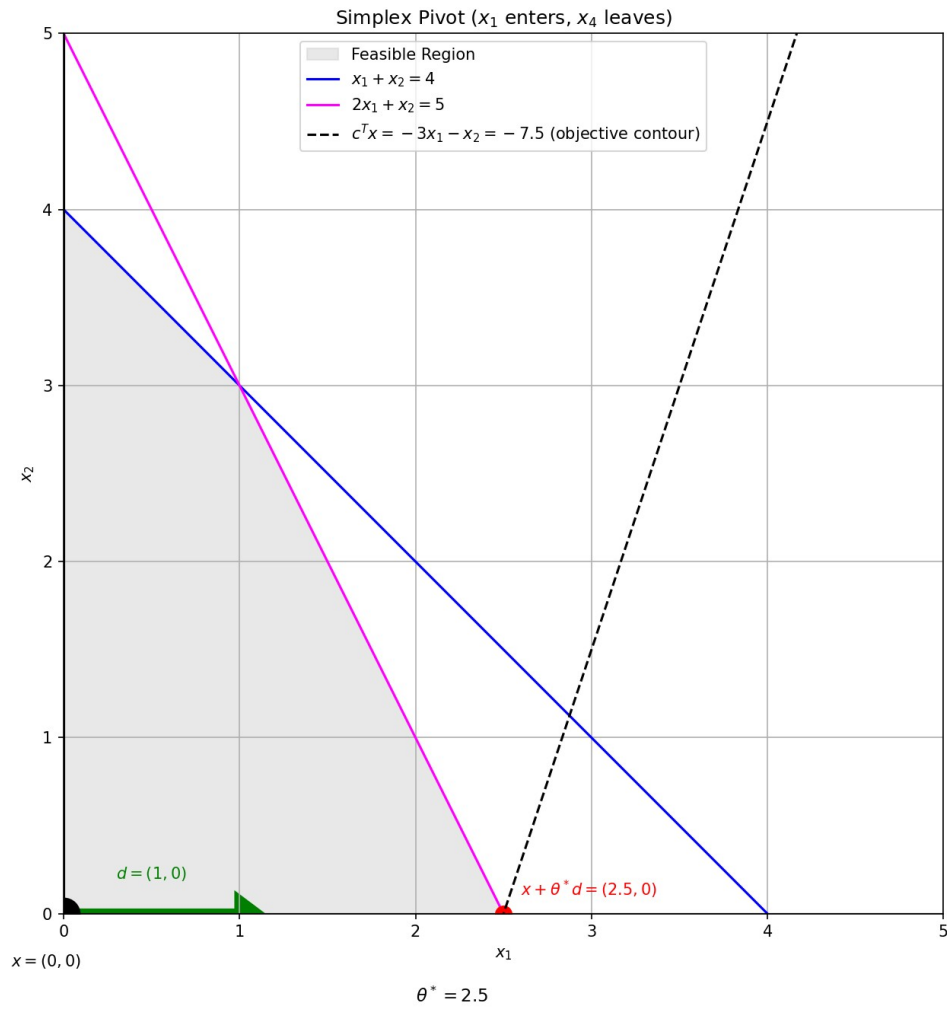


Figure 1: Visualization of the pivot where the non-basic variables x_1 becomes basic (enters the basis) and the basic variable x_4 becomes non-basic (leaves the basis).

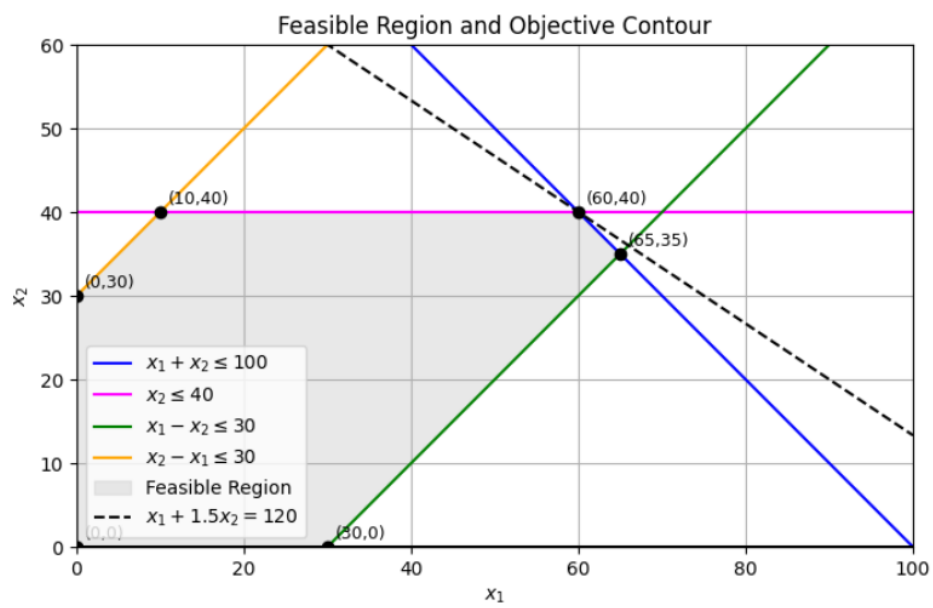


Figure 2: Visualization of the feasible region and objective contour of the practice exercise.